GuardianEye

Multimodal Late Fusion of Image and Audio Poaching Indicators

Multimodal AI (MAI) · Group 27 · Academic Year 2026–2027

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Problem Formulation

Why multimodal AI for wildlife indicators

- Patrol evidence arrives as RGB stills, aerial thermal frames, and short audio clips.
- Streams are rarely time-aligned; a single sensor reading is incomplete.
- No labelled poacher class in the configured collections.
- One detection is an indicator, not proof of poaching.
- Video is sampled into frames and follows the image branch — not a third modality.
- Late fusion combines unimodal scores at decision time, not in a joint encoder.

Objectives

Completed multimodal work only

- Train separate visual detectors for aerial_thermal and camera_trap_rgb (YOLOv8).
- Classify gunshot vs background on log-mel windows with a convolutional network.
- Evaluate audio under Belize↔️Vietnam transfer and in-domain controls.
- Fuse modalities with rules, time windows, and missing-modality degradation.
- Build synthetic proxy pairs when field-aligned pairs do not exist.
- Serve a multimodal API and Gradio dashboard with an honest audit log.

Multimodal Architecture

Two unimodal branches + late fusion

Image branch

- YOLOv8 per domain
- human / vehicle / animal / empty
- Empty frames retained

Audio branch

- 16 kHz · 3 s windows
- 64-bin log-mel CNN
- Class-weighted loss

Late fusion

- Rules mode (primary)
- 5 s coincidence window
- HIGH / MEDIUM / LOW

Visual Modality

Domain-separated object detection

- Independent YOLOv8 runs for aerial_thermal and camera_trap_rgb.
- Thermal uses channel replicate into three identical channels for an RGB backbone.
- Rare classes oversampled; empty negatives kept.
- Visual indicators for fusion: human and vehicle only.

Domain	Human	Vehicle	Animal	Empty	Total
aerial_thermal	2	1	2	6	11
camera_trap_rgb	2	1	2	6	11

Audio Modality

Site-stratified acoustic classification

- Log-mel CNN classifies gunshot vs background.
- Background and distractor impulses retained.
- Cross-site experiments measure geographic transfer into fusion.

Site	Gunshot	Background	Total
Belize	4	10	14
Vietnam	5	10	15

Cross-Site Audio Experiments

Transfer both ways + in-domain controls

Experiment	Train	Validation	Test
belize_to_vietnam	11	3	14
vietnam_to_belize	11	3	14
in_domain_belize	8	3	3
in_domain_vietnam	8	3	3

Validation stays on the training site. Cross-site tests use the other geography entirely. Fusion inherits whatever site shift the audio encoder exhibits.

Late Fusion Policy

configs/fusion.yaml · rules mode

- HIGH — human/vehicle and gunshot, both ≥ 0.50, inside a 5 s window → default alert.
- MEDIUM — exactly one strong indicator, or both outside the window.
- LOW — neither indicator.
- Missing modality: unimodal thresholds 0.85; level cannot be HIGH.
- Weight profiles (balanced / night / wind) available for weighted scoring.
- Animal detections are logged and excluded from poaching indicators.

Synthetic Proxy Multimodal Pairs

Not field-aligned incidents

- 16 constructed pairs.
- Protocol tag: synthetic_proxy on every row.
- Eight scenarios × two visual domains.
- Used to compare image-only, audio-only, and fused decisions.
- Explicitly labelled as a proxy protocol in reports.

Expected level	Pairs
HIGH	6
MEDIUM	6
LOW	4

Multimodal Serving

FastAPI + Gradio + SQLite audit

- API on port 8742; dashboard at /dashboard accepts image and audio together.
- GET /health — process liveness.
- GET /model-info — per-branch checkpoint honesty (loaded vs none).
- Every decision logged to data/events.sqlite, including non-alerts.
- Banner: proof-of-concept. Indicators, not confirmed poaching.
- Current tree: no checkpoints in models/; evidence marked unreliable.

Multimodal Corpus Snapshot

Metadata counts only — not detection scores

Quantity	Value
Image records	22
Empty images retained	12
Audio windows	29
Background clips retained	20
Audio split assignment rows	84
Synthetic proxy pairs	16
Checkpoints in models/	none

Conclusion

Multimodal AI deliverable

- GuardianEye implements late multimodal fusion of vision and audio indicators.
- Unimodal branches stay specialised; fusion is rules-based and time-aware.
- Site stratification and empty/background retention keep evaluation honest.
- Synthetic proxy pairs stand in where field-aligned multimodal data do not exist.
- Serving exposes multimodal uploads with explicit reliability reporting.
- No fabricated mAP, F1, precision, or recall — metrics require checkpoints.

Thank you

Questions welcome

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